

ML_PCOM7E April 2026

Unit 6: Development Team Project

Airbnb Business Analysis Using a Data Science Approach

Track 2: Deep Learning (Neural Networks, CNNs)

Module: Machine Learning

Assignment Component: Part 1 – Design Proposal Document

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1. Introduction

Airbnb operates in a competitive marketplace where identifying listings with stronger guest activity can support host guidance, pricing advice, search ranking, and neighbourhood-level planning. This report investigates whether deep learning can help identify listings with stronger guest activity in New York City using the data available in the AB_NYC_2019 dataset (Kaggle, n.d.).

2. Business Question

The proposed business question is: Can Airbnb use deep learning to identify high-demand NYC listings and support host guidance using structured, text-based, and spatial listing data?

3. Methodology

3.1. Target definition and proxy validation

Because the dataset lacks direct demand measures such as bookings or occupancy, `reviews_per_month` was used as a proxy for guest activity, with the top 25% of listings flagged as `high_demand`. A high number of positive reviews is a reported key influence driving consumer trust (Holiday Travel, n.d.), and could therefore be considered a conversion factor for bookings. This proxy is measurable and reproducible, and Figure 1 suggests it is not random as review activity varies by location and listing-title length. These patterns support further spatial and text-based analysis, with prior research also linking property-title language to rental performance (Heo and Park, 2024).

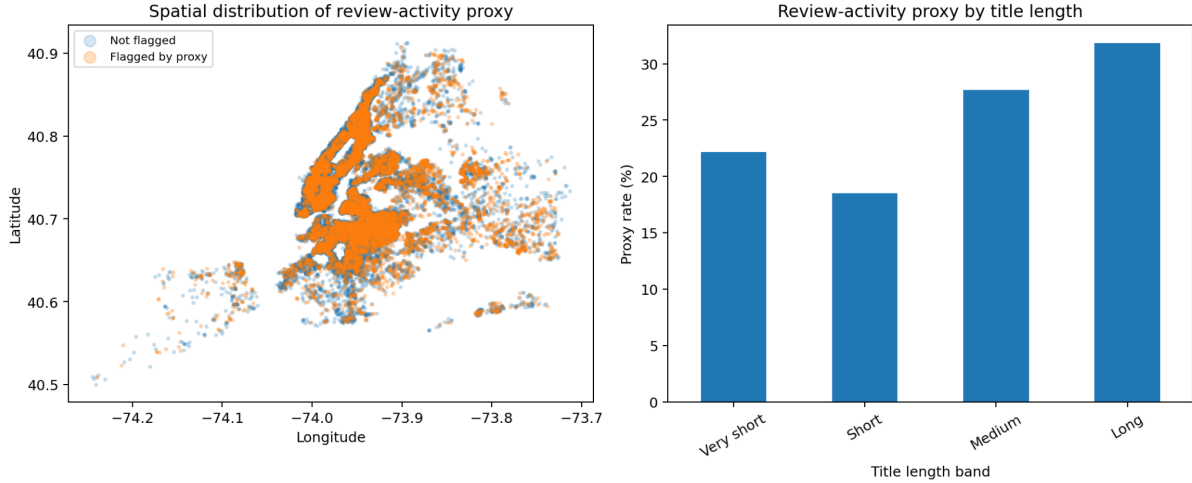


Figure 1: Spatial and title-length patterns in the review-activity proxy.

3.2. Tabular neural network

A neural network (NN) was trained on the structured baseline features shown in Table 1, using the derived binary `high_demand` target. Before modelling, zero-price listings were removed, skewed features such as `price` and `host_listings` were log-transformed, `minimum_nights` was capped at the 99th percentile and categorical values were encoded. Multicollinearity was examined to optimise accuracy and model stability. Finally, data was split into train/test subsets and standardised.

Baseline models served as benchmarks to determine if the NN could add value beyond simpler approaches (Table 2). The NN was optimised using `class_weights` and Early Stopping to address the 75:25 class imbalance and reduce overfitting.

Features	Status	Rationale
id, host-id, host_name	Excluded	Do not provide predictive value for future/new listings
name	Excluded (used for text/NLP)	Utilised for text-based model
neighbourhood_group, latitude, longitude	Retained	Captures geographical variance
neighbourhood	Excluded	High number of categorical values and redundancy with other geographical features
price, minimum_nights, availability_365	Retained	Captures listing variables that may optimise demand
last_review, number_of_reviews	Excluded	Target leakage
calculated_host_listings_count	Retained	Captures role of professional hosts in guest activity

Table 1: Feature selection.

Model	Accuracy	ROC-AUC	Recall 0	Recall 1
Logistic Regression	74%	0.738	0.85	0.25
Random Forest	81%	0.853	0.88	0.59

Table 2: Model comparison results.

3.3. Sentiment analysis

The sentiment model used listing text, primarily the name field, to capture tone and presentation signals in Airbnb listings. An LSTM was selected because it can retain information across word sequences more effectively than a standard RNN, making it suitable for text-based sentiment classification (Sherstinsky, 2020; Van Houdt, Mosquera and Nápoles, 2020).

3.4. CNN map-patch

A CNN model converted each listing's latitude and longitude into a 32×32 grayscale map-patch from an NYC map. These patches were used as image inputs to predict whether a listing was in high demand.

3.5. Hybrid network

Finally, the hybrid network combined structured tabular inputs with continuous probability outputs from the LSTM and CNN models: `sentiment_prob` and `cnn_demand_prob`. These auxiliary features were added to the final classifier to predict `high_demand` (Figure 2).

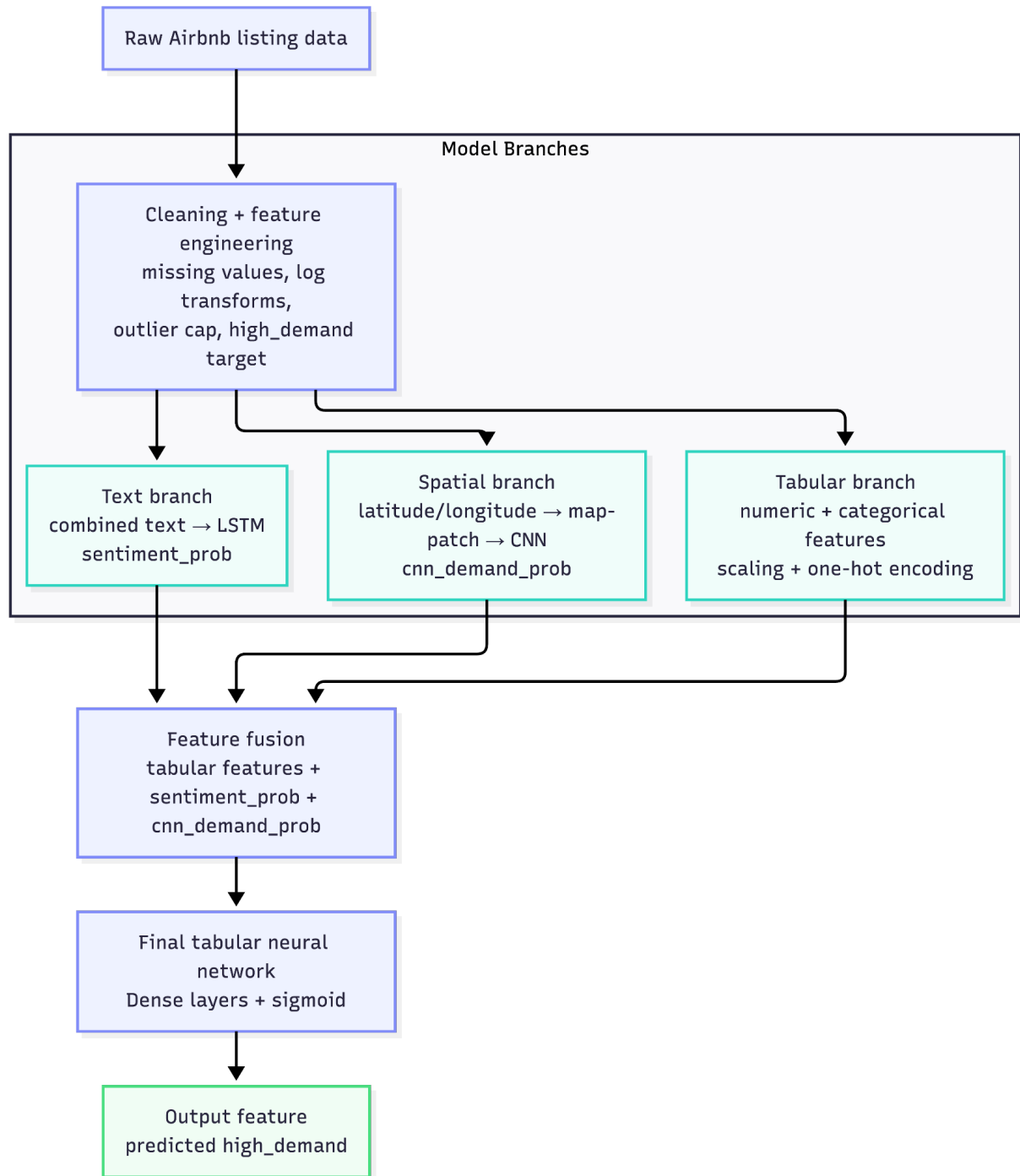


Figure 2: Hybrid model architecture.

4. Analysis and Results

Model	Accuracy	ROC-AUC	Recall		Precision		F1	
			0	1	0	1	0	1
Tabular NN	0.75	0.83	0.75	0.76	0.90	0.50	0.82	0.60
Sentiment Analysis	0.96	-	-	0.96	-	0.95	-	0.96
CNN Map-patch	0.51	0.52	0.50	0.54	0.76	0.26	0.60	0.35
Hybrid Network	0.79	0.83	0.90	0.46	0.83	0.62	0.87	0.53

Table 3: Model performance summary.

4.1. Tabular NN

The optimised tabular NN achieved strong performance (Table 3), with permutation importance identifying `availability_365` and `minimum_nights` as key predictors of high demand (Figure 3).

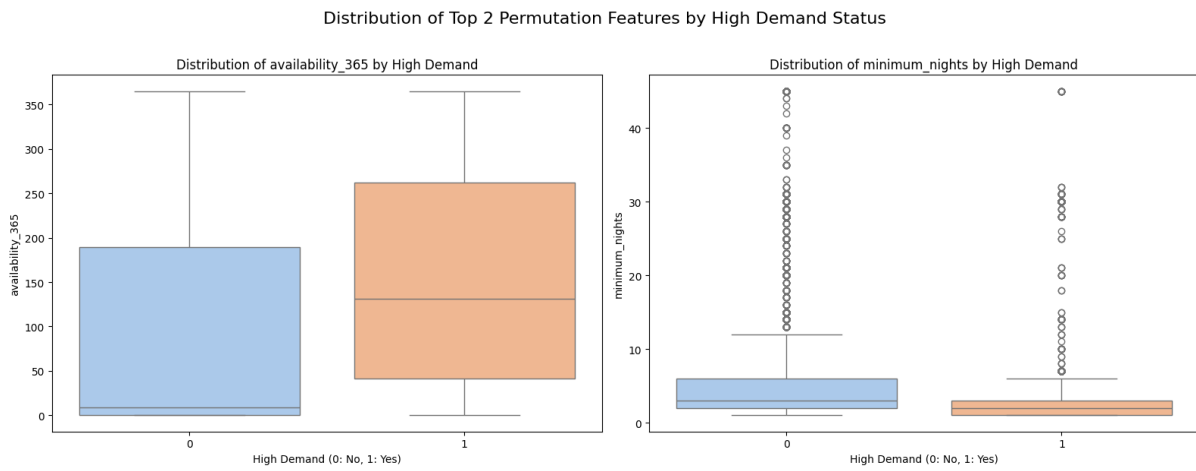
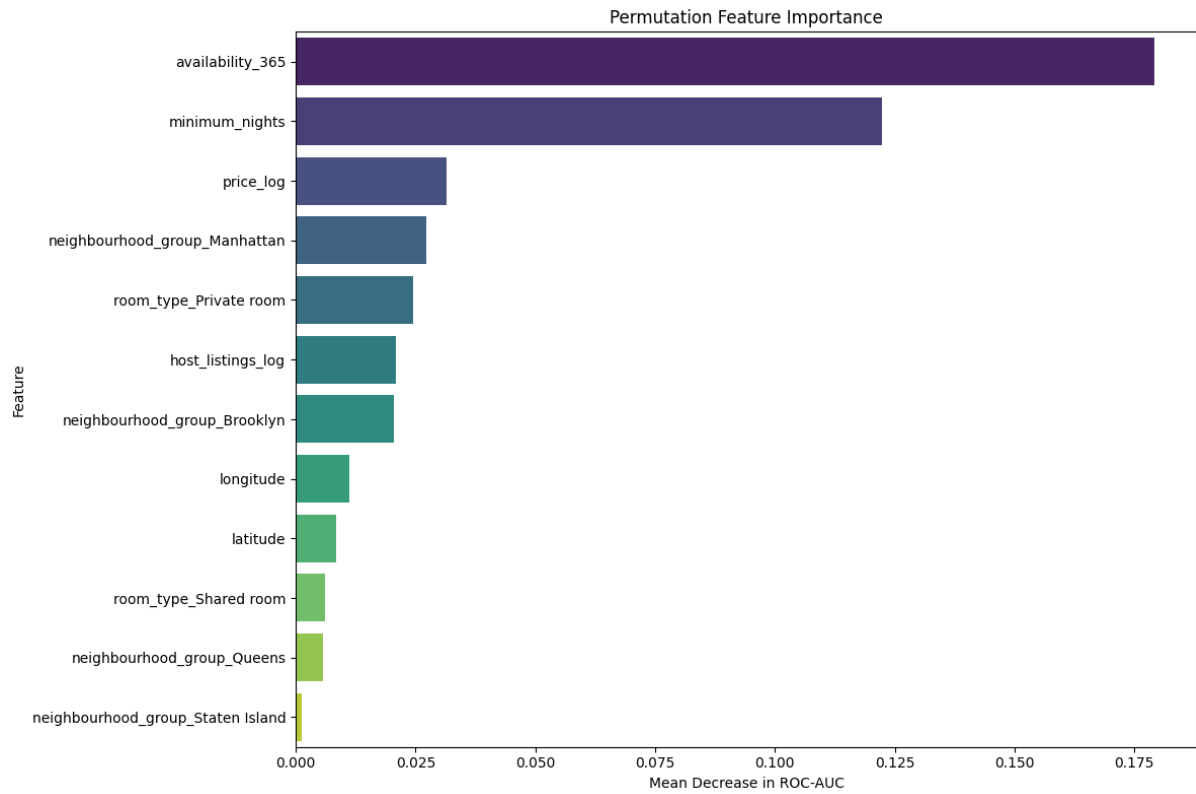


Figure 3: Tabular NN feature importance and top feature distributions.

4.2. Sentiment analysis

The sentiment model learned effectively, with training loss falling across epochs (Figure 4). As shown in Table 3, it achieved strong classification performance across accuracy, precision, recall, and F1-score. Positive Airbnb sentiment in NYC was associated with popular neighbourhoods and appealing housing features like spacious, sunny, and cozy apartments (Figure 5).

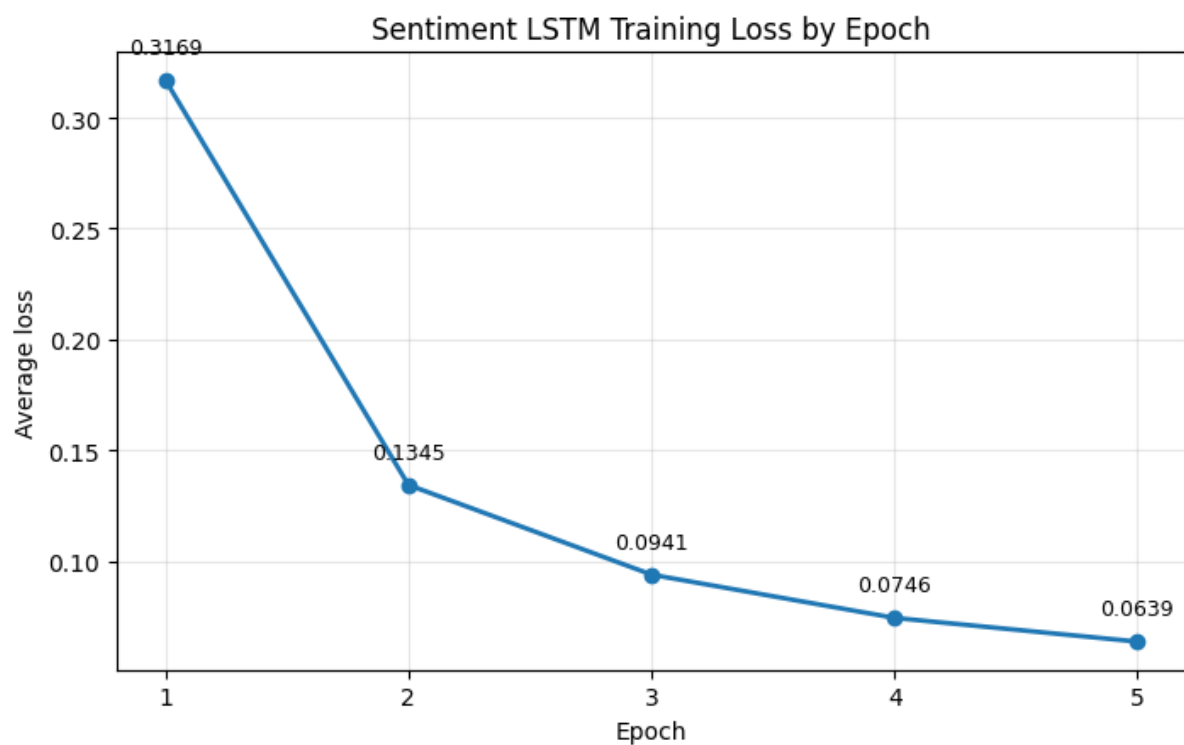


Figure 4: LSTM training loss by epoch.

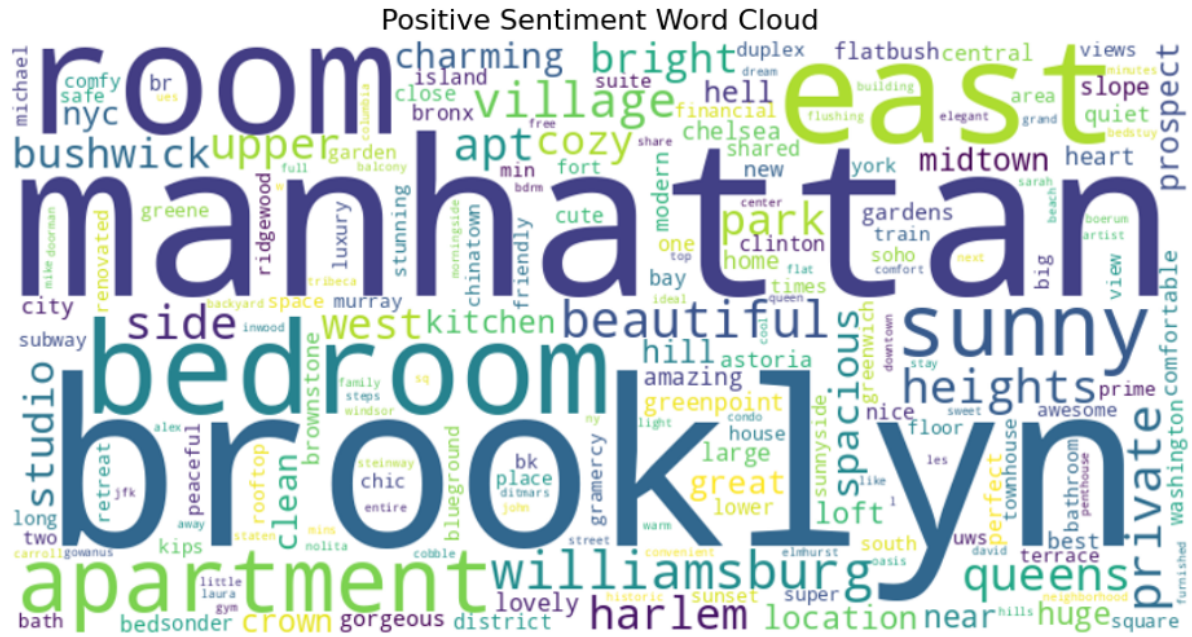


Figure 5: Positive Sentiment Word Cloud.

4.3. CNN

The CNN map-patch model performed weakly (0.51 accuracy and 0.52 ROC-AUC). Although it detected some spatial/edge patterns, the high- and low-demand patches were visually similar (Figure 6), causing poor class separation and many false positives.

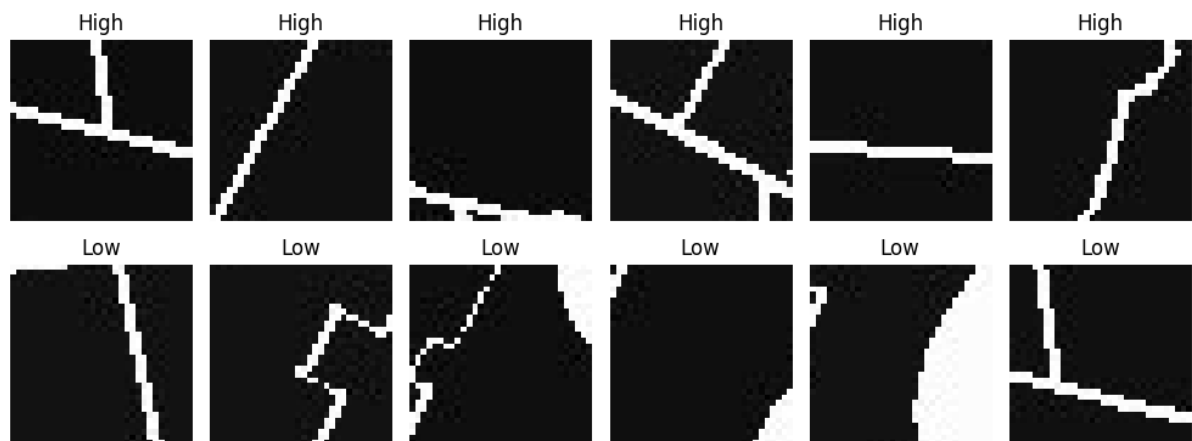


Figure 6: CNN map-patch outputs.

4.4. Hybrid model

The hybrid network achieved 79.3% accuracy and 0.833 ROC-AUC, but high-demand recall fell to 0.46, suggesting it was cautious and missed more high-demand listings. However, the score-band analysis shows predicted demand scores could still support different business actions at different priority levels (Figure 7).

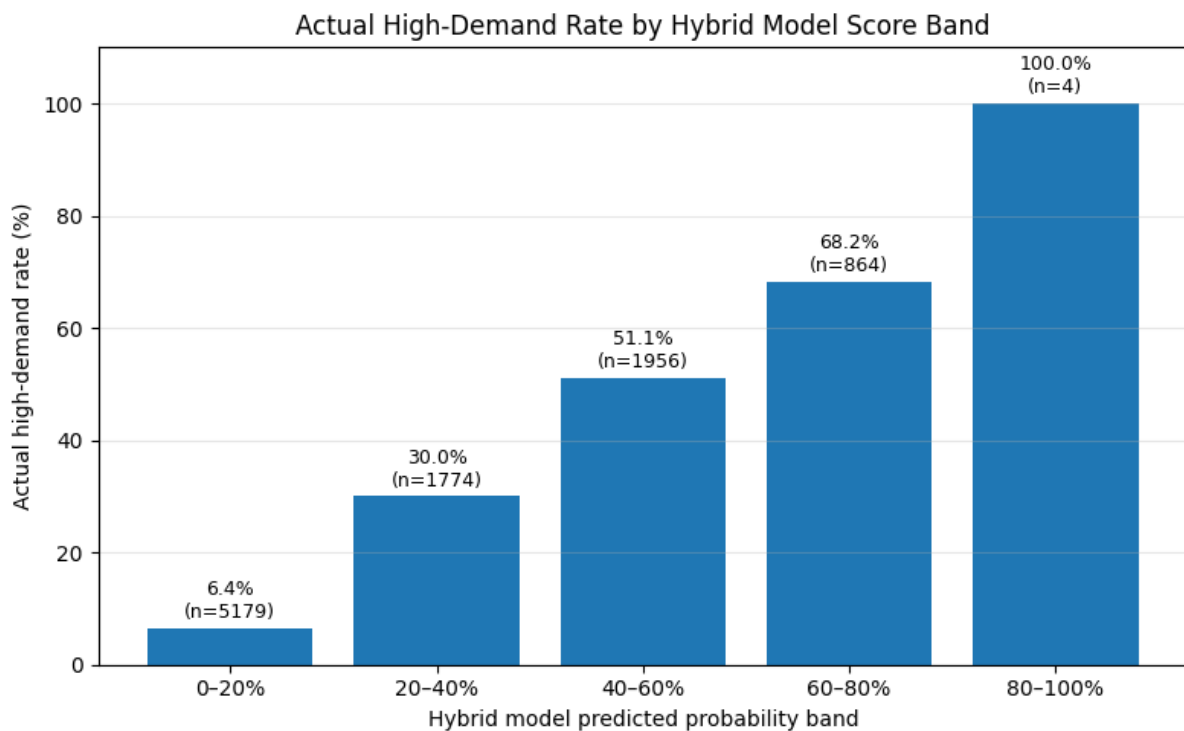


Figure 7: Actual high-demand rate by hybrid model score band.

5. Discussion

5.1. Model interpretation

The tabular neural network provided the strongest business insight, identifying `availability_365` and `minimum_nights` as important predictors of `high_demand`, suggesting hosts may improve guest activity by maintaining stronger availability and avoiding unnecessarily high minimum-stay requirements.

The sentiment model performed strongly, suggesting that listing text captured useful tone and presentation signals. However, the CNN map-patch model added limited value, indicating that simplified map images were less informative than structured location features.

The hybrid model did not clearly improve over the tabular baseline. Its performance may have been affected by weak CNN output, which may have added noise rather than useful additional information. Even so, its demand score bands may still provide a useful priority signal for business workflows, such as host guidance, listing review prioritisation, and search-ranking support. For example, listings in high-score bands could be prioritised for promotion, while mid-score listings could receive host guidance around availability, `minimum_nights`, and listing presentation.

5.2. Limitations

A key limitation is that the dataset did not contain variables directly capturing occupancy. Using the demand proxy relies on several assumptions, including that all guests leave reviews, reviews are positive and stay duration is consistent.

Utilisation of an NN to analyse the dataset has several advantages including the ability to map complex non-linear relationships and incorporate unstructured data (Goodfellow et al., 2016; LeCun et al., 2015). Despite this, research shows that gradient boosted decision trees often outperform NNs on tabular data (Grinsztajn et al., 2022). Additionally, NNs lack inherent interpretability regarding feature importance, complicating interpretation for business insight (Guidotti et al., 2018).

The sentiment model measured sentiment labels derived from listing text rather than demand itself. Its strong performance shows that it learned tone and presentation

patterns, but not necessarily whether those listings generated more bookings or occupancy.

6. Conclusion

This project found that Airbnb can use deep learning to identify higher-demand NYC listings and support host guidance, with strongest value from structured listing data.

Hosts should maintain strong availability and avoid high minimum-night requirements, as these were the clearest actionable demand signals. Listing text also showed useful presentation patterns, suggesting clearer titles may support guest activity. Although the hybrid model did not outperform the tabular baseline, model score bands from either approach could support promotion decisions by helping Airbnb prioritise listings with stronger guest activity.

Overall, deep learning can support host guidance, listing prioritisation, and promotion workflows, but richer booking, occupancy, revenue, or review data is needed for commercially reliable recommendations.

References

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